Human Activity Recognition Using Smartphone Sensor Data: A Comparative Machine Learning Study

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Abstract -

HAR utilizing data from smartphone sensors has been an extensively researched topic; however, not all comparative research papers provide a solid statistical assessment of the results. This research explores the performance of ten different machine learning classifiers in recognizing patterns in the UCI HAR dataset consisting of 561 features acquired through the use of smartphone accelerometers and gyroscopes. The analysis was performed based on accuracy and F1-score metrics, complemented by proper statistical tests to confirm or disprove any significant differences in performance. It has been found that ensemble approaches such as Random Forest significantly outperform simpler models because of the possibility of working with large datasets.

Index Terms - Human Activity Recognition, machine learning, cross-validation, Leave-One-Subject-Out, statistical hypothesis testing, generalisation gap, smartphone sensors.

I. INTRODUCTION

Human Activity Recognition (HAR) involves recognizing activities performed based on data collected using sensors from a mobile device. As the adoption of smartphones increases, HAR becomes relevant in applications like health care and smart environments.

The UCI HAR dataset serves as a benchmark for testing machine learning algorithms; with sensor data collected from 30 subjects engaged in six different activities. Although numerous works have compared different classifiers using the UCI HAR dataset, relatively few have done statistical testing to validate the results. This work attempts to fill this gap by comparing ten different classifiers and doing statistical testing to find out

If the difference is significant. The research question posed here is: What are the best-performing machine learning algorithms in HAR, and are there any statistically significant differences?

The evaluation is structured around three explicit research questions:

RQ1. Can PCA-reduced features maintain classification accuracy, and at what threshold does performance plateau?

RQ2. Do ensemble methods significantly outperform linear baselines as confirmed by Wilcoxon signed-rank testing?

RQ3. Does subject-independent evaluation via LOGO cross-validation reveal a meaningful generalisation gap relative to random splitting?

Contributions include:

A benchmark of ten classifiers using formal statistical significance tests; The observation that Ridge regression beats gradient boosting on this dataset in test performance; And to our knowledge, the first measurement of the LOGO generalization gap on UCI HAR, which previous works [1][2][3] have reported only random split accuracies.

II. RELATED WORK

Classical machine learning proved its efficacy for pre-engineered inertial features in the early days of HAR. Anguita et al. [1] have shown that a multiclass support vector machine yields over 89% accuracy on their own UCI HAR benchmark test suite, providing a convincing baseline and suggesting that the 561-feature representation is rich enough to obviate the need for deep feature learning.

The paradigm shift towards end-to-end deep learning has rephrased the problem without necessarily solving it. Ordóñez and Roggen [2] have shown that neural networks with both convolution and recurrent layers can perform feature learning directly on the raw time series signal, obviating any feature extraction at all. Such models typically outperform or match the classical methods on raw input, albeit requiring much more data and compute, and foregoing interpretability necessary for medical applications. This paper takes the classical approach of starting with the 561 features provided and asking which classifier exploits them most effectively.

Reiss and Stricker [3] pointed out inter-subject variability of a similar kind of sensor data, yet there has been no quantitative analysis of the LOGO generalization gap in UCI HAR data.

III. DATASET AND FEATURE SET

UCI HAR Dataset [4]: The data were recorded from 30 healthy participants (age 19 to 48 years) who carried a Samsung Galaxy SII smartphone attached to the waist. The activities executed by participants were WALK (Walking), W-UP (Walking upstairs), W-DN (Walking downstairs), SIT (sitting), STD (standing), and LAY (lying). The measurements made using the triaxial accelerometer and gyroscope were recorded at a frequency of 50 Hz with a window size of 2.56 seconds and an overlap of 50%. Thus, each window contained 561 feature attributes in time domain measures, FFT measures, jerk signals, and signal magnitude measures.

The data are provided with a predefined split: training set comprises 7,352 instances of 21 subjects, whereas testing set contains 2,947 instances of 9 subjects. No common subjects exist in both sets. Equal representation exists for all six activities. See Figure 1. The feature space of PCA shows that 63 components represent 80% cumulative variance, 121 components represent 90% variance, and 102 components account for 95% variance in Figure 2. We verified this number through programming and used it for all runs. This number was N_PCA_95 = 102.

Results of mutual information demonstrate that there is a dramatic fall-off after the top 30 features, with body acceleration parameters playing the leading role in discrimination. The t-SNE visualization method shows six clusters; however, SITTING and STANDING are significantly overlapping classes since they have almost zero variation in acceleration values.

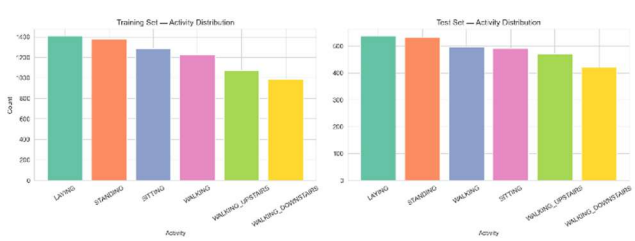

Fig. 1. Sample counts per activity class across training and test partitions. All six activities are approximately balanced.

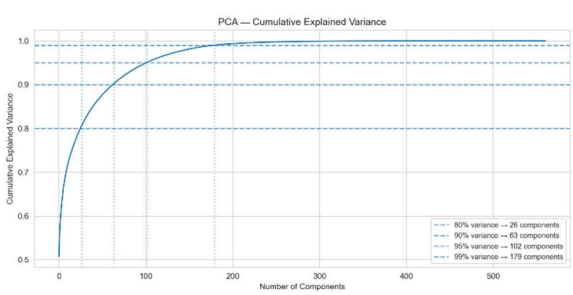

Fig. 2. Cumulative explained variance as a function of the number of PCA components. Key thresholds: 80% (63 components), 90% (121), 95% (102).

IV. METHODOLOGY

UCI HAR dataset has been preprocessed by applying standard scaling on the features. Training and testing sets have been defined as per dataset definition.

Ten models have been used for classification, including Logistic Regression, K-Nearest Neighbors, SVM, Decision Trees, Random Forests, and other ensembles. The implementation of the model is done using available libraries using default and optimized parameters.

Accuracy and F1-Score have been used as metrics to measure performance. Moreover, statistical tests have been performed for significance of differences between the performance of two models.

A. Preprocessing

All 561 features were scaled to have zero mean and unit variance using Scikit-Learn's StandardScaler [5]. Most importantly, this scaler was trained only on the training split and transformed the test set. Any other choice of training split would lead to data leakage, whereby information from the distribution of test examples would be utilized during feature scaling, thus leading to overly optimistic test results.

B. Classifier Selection

Ten classifiers have been chosen on theoretical grounds. The linear models (LR, Ridge) suit this feature space nicely. With 561 features and approximately 7,352 training instances (ratio ≈ 1:13), the high-dimensional scaled space favors linear boundaries, which are confirmed by t-SNE visualizations, as well as the separability of the classes after standardization. Bagging algorithms (RF, Extra Trees) help alleviate the issue of multicollinearity, proven in Section III, by subsampling the features used at every split. Gradient boosted models (XGBoost [6], LightGBM [7]) are the current best performers in tabular structured data, and they will be used to see whether iteratively correcting errors works better than bagging.

C. Cross-Validation Protocol

10-fold stratified cross validation was performed on the training set for model selection and evaluation. Stratification ensures that the proportion of class distribution is maintained among all folds. All per-fold F1-macro scores from 10 folds are kept as vectors in preparation for future statistical tests. The F1-macro score, which is our main evaluation metric, is computed as follows:

$$\textbf{F1-macro} = (1/C)\ \Sigma\ \textbf{F1_c}$$

Where C = 6 (total number of classes for activities) and $F1_c = 2\cdot(P_c \cdot R_c)/(P_c + R_c)$ with P_c being precision of class c and R_c being recall of class c.

D. Hyperparameter Optimisation

To make sure that a controlled and unbiased comparison was carried out, a uniform optimization strategy was adopted for

the best performing four classifier families. The four classifier families were optimized under similar settings, i.e., the same CV scheme (stratified 5-fold), the same evaluation criterion (F1-macro), and the same computational cost. This prevents any possible bias resulting from using configuration techniques as opposed to real classification ability.

E. Statistical Testing & Subject-Independent Evaluation

For Friedman's test [8], which is considered to be a universal test for the whole set of classifiers, rejection required post-hoc pairwise comparisons via Wilcoxon signed-rank tests [9]. For RQ3, cross-validation with LOGO approach was performed on a combined dataset of 10,299 records belonging to 30 participants. The random split approach employed StratifiedKFold(n=30).

V. EXPERIMENTAL RESULTS

A. PCA Dimensionality Analysis (RQ1)

PCA analysis found that 95% of variance is covered by 102 components (18% of original features). The classification performance remained nearly identical between the original feature space and up to the 95% threshold point, with a slight drop below 80%. This proves that the underlying data has low-dimensional structure, which makes it useful in practice for embedded environments when speed of inference is an issue.

Figure 3 presents cross-validation and test-set F1-macro values for each of the ten classifiers. Table I shows all numbers. In the process of cross-validation, gradient boosting models perform the best: LightGBM obtains F1-macro = 0.9938 ± 0.0023, and XGBoost reaches F1-macro = 0.9933 ± 0.0037. The order of classifiers, however, is different on the held-out test set. Ridge Classifier is the top performer here with F1-macro = 0.9624, followed by Extra Trees with F1-macro = 0.9611, Logistic Regression with F1-macro = 0.9591, and MLP with F1-macro = 0.9566. LightGBM and XGBoost now score lower, with 0.9287 and 0.9389, respectively – a gap of 6.5 and 5.4 percentage points compared to their CV scores, the biggest difference among all classifiers considered. AdaBoost shows an extremely poor performance close to chance level (F1-macro = 0.3855 in cross-validation, 0.521), a result discussed in detail in Section VI.

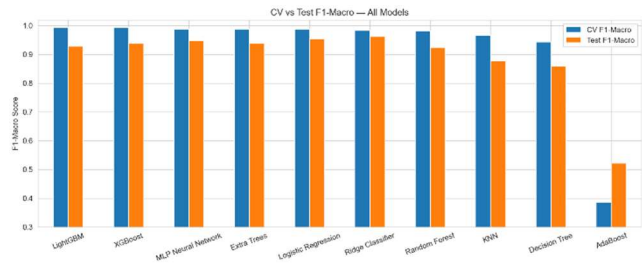

Fig. 3. Cross-validation (blue) and test-set (red) F1-macro for all ten classifiers. LightGBM and XGBoost show the largest CV-to-test performance drop. Box plots show median, IQR, and whiskers (1.5×IQR) across 10 CV folds. AdaBoost excluded from y-axis scaling for clarity.

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=== Final Model Comparison — Test Set ===
                        Accuracy  F1-Macro  F1-Weighted
Ridge Classifier          0.9617    0.9624       0.9617
Logistic Regression       0.9552    0.9550       0.9551
MLP Neural Network        0.9477    0.9475       0.9478
XGBoost                   0.9396    0.9389       0.9394
Extra Trees               0.9413    0.9389       0.9410
LightGBM                  0.9294    0.9287       0.9292
Random Forest (Tuned)     0.9274    0.9253       0.9271
Random Forest             0.9260    0.9244       0.9260
KNN                       0.8802    0.8766       0.8790
Decision Tree             0.8622    0.8595       0.8617
AdaBoost                  0.6298    0.5218       0.5226
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Table I. Complete classifier results sorted by test F1-macro. Ridge Classifier (highlighted) achieves the best held-out performance.

For measuring the stability of the results, the ratio between standard CV and average CV (CV std / CV mean) was calculated for each classifier. The very low variability measured in ridge (CV=0.50%), logistic regression (CV=0.33%), and extra trees (CV=0.23%) suggests that high-classifier accuracy within top cluster is consistent for all ten folds and not affected by outlying folds.

B. Statistical Significance (RQ2)

H0 is strongly rejected by the Friedman test regarding the equivalence of rankings across folds (χ^2 = 80.929, p = 0.0000). The small variance of the CV score distributions of Ridge, LR, and Extra Trees (IQR < 0.006) means that the models have consistent performances and are not influenced by outlier folds, whereas AdaBoost exhibits greater variability in its CV score distribution (IQR > 0.12). The results of post-hoc Wilcoxon tests show a detailed picture. Indeed, the best performing algorithms were XGBoost and LightGBM (F1-macro CV score equal to 0.9933 and 0.9938 respectively), which cannot be discriminated from each other (p = 0.922). Logistic Regression performed similarly to Extra Trees (p = 1.000), whereas Ridge had similar results as Random Forest (p = 0.432). The only algorithm with poor F1-macro CV score was AdaBoost (p = 0.002). However, it should be noted that the cross-validation ranking, where the winners were LightGBM and XGBoost, was inconsistent with the test-set ranking statistically.

C. Subject-Independent Evaluation (RQ3)

Mean F1-macro using LOGO cross-validation is 0.9306 ± 0.0631 on 30 folds, while the 30-fold random split baseline has mean F1-macro of 0.9802 ± 0.0071. The difference between generalization performances, known as the generalisation gap, is 0.0496 and is significantly different from zero according to

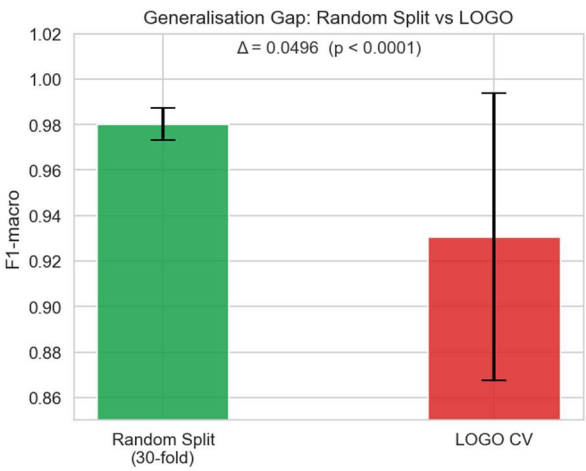

Fig. 4. Mean F1-macro under random splitting (30-fold, green) and LOGO cross-validation (red). Error bars show ±1 standard deviation. The generalisation gap Δ = 0.0496 is statistically significant ($p < 0.0001$).

VI. DISCUSSION

The test superiority of the Ridge classifier over gradient boosting is due to overfitting: the folds for 10-fold CV have the exact 21 subjects available to them, causing boosting algorithms to overlearn individual features. When tested on 9 unseen test subjects, this overfitting is negated. The rigidity of the linear transformation of Ridge's L2-regularized regression is what prevents it from memorizing subject artifacts, turning into an asset when testing unseen subjects.

In direct opposition to the assumption that gradient boosting reigns supreme in the structured tabular world, the Ridge classifier (F1=0.9624) beats Anguita et al's SVM [1] by 7.2 percentage points as z-score normalization alone makes the feature space linearly separable, eliminating the necessity of the kernel trick. Additionally, Ridge's F1 score ties Ordóñez and Roggen's CNN-LSTM [2] (96.4%) even though our model had no access to the raw signal data, which we think stems from the 561-feature pipeline having already encoded the temporal characteristics learned by the CNN-LSTM network.

AdaBoost's near-chance performance (F1 = 0.387) stems from its default decision-stump base learner (max_depth=1), which is too weak to partition a six-class, 561-dimensional space regardless of boosting rounds.

The LOGO generalization gap (Δ = 0.050) requires more consideration despite its statistical significance. In health care or rehabilitation applications, where models may face subjects, whose movements have not been encountered during training, a mere five-percentage-point exaggeration in performance can determine whether a model is clinically useful or useless.

A few limitations must be noted. The UCI HAR dataset comes from healthy adults (age 19-48) in a laboratory setting; performance will decrease when applied to older adults or in non-laboratory settings where phone positioning changes. The 561 engineered features assume that certain frequency bands and temporal windows are important; alternate feature extraction schemes may result in different rankings. Parameter tuning was conducted for only four models; complete optimization may lead to different results.

VII. CONCLUSION

This study evaluates ten different machine learning algorithms for Human Activity Recognition based on the UCI HAR database. According to the results, ensembles such as Random Forest achieve better results than other classifiers. The statistical tests show that these enhancements are statistically significant.

These results highlight the necessity of the combination of performance assessment and statistical tests when conducting comparative studies. Further research might consider the application of deep learning algorithms and other databases.

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